

CLASSROOM CONTACT PROGRAMME

(ACADEMIC SESSION 2019-2020)

PRE-MEDICAL : ENTHUSIAST COURSE [PHASE-II]

TEST # 05

DATE : 12-01-2020

REVISION TEST

Test Pattern : NEET

INSTRUCTIONS

Do not open this Test Booklet until you are asked to do so

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **180** questions. The maximum marks are **720**.
5. There are **three** parts in the question paper A,B,C consisting of **Physics, Chemistry** and **Biology** having **45 questions** each in Physics & Chemistry and **90 questions** in Biology part of equal weightage. Each question is allotted 4 (four) marks for **correct** response.
6. **One Fourth** mark will be deducted for indicated incorrect response of each question. **No deduction** from the total score will be made if no response is indicated for an item in the Answer Sheet.
7. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** the Answer Sheet. **Use of pencil is strictly prohibited.**
8. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
9. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
10. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
11. **Do not fold or make any stray marks on the Answer Sheet.**

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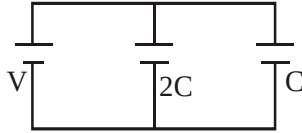
Your Target is to secure Good Rank in NEET 2020

SPACE FOR ROUGH WORK

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

1. Two condensers, one of capacity C and the other capacity $2C$, are connected to a V volt battery, as shown in figure

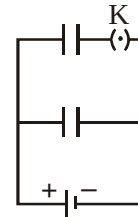


The work done in charging fully both the condenser is

- (1) $\frac{1}{2}CV^2$ (2) $2CV^2$
(3) $\frac{3}{2}CV^2$ (4) $3CV^2$
2. The energy required to charge a parallel plate condenser of plate separation d and plate area of cross-section A , such that the surface charge density on the plate will be σ , is
- (1) $\frac{A\sigma^2 d}{2\epsilon}$ (2) $\frac{A\sigma^2 d}{\epsilon}$
(3) $\frac{\epsilon\sigma^2 d}{2}$ (4) $\frac{A\epsilon\sigma^2}{d}$
3. Three identical capacitors are connected in series. The capacitance of combination is C and break-down voltage is V . The net capacitance and break down voltage when these are connected in parallel combination will be :

- (1) $\frac{C}{3}, 3V$ (2) $9C, \frac{V}{3}$
(3) $\frac{C}{3}, V$ (4) $3C, \frac{V}{3}$

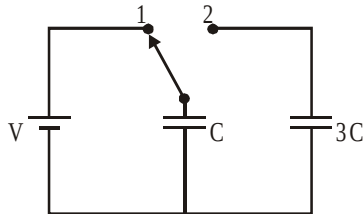
4. Two identical parallel plate capacitors are connected as shown in the circuit. The key K is unplugged and identical dielectric material is filled in both capacitors. Choose the correct statement.



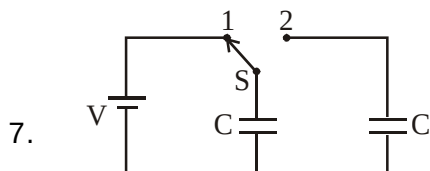
- (1) Both capacitor will have same charge
(2) Both capacitor will have same potential
(3) Both capacitor will have same energy stored.
(4) The ratio of energy stored in capacitors will be $1:K^2$.
5. A parallel plate capacitor has uniform electric field E between the plates and surface area of each plate is A . The force of attraction between the plates of the capacitor is
- (1) $E^2 A \epsilon_0$
(2) $2E^2 A \epsilon_0$
(3) $\frac{E^2 A \epsilon_0}{2}$
(4) $\frac{E^2}{2A \epsilon_0}$

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6. Two capacitors of capacitance C and $3C$ are shown in the diagram. When switch S turned to position 2, the percentage of stored energy dissipated is



- (1) 0% (2) 25%
(3) 75% (4) 100%



Two identical capacitors as shown in the diagram. when switch S is turned to position 2, then choose incorrect statement :-

- (1) Charge on both capacitor will be same
(2) Potential difference across each capacitor will be half
(3) Energy stored in capacitors remain unchanged.
(4) Energy stored in capacitors will be reduced to half.

8. Select incorrect statement for the electrostatic force between the metal plates of an isolated charged parallel plate capacitor :

- (1) Inversely proportional to the area of plates
(2) Directly proportional to square of charge on the plates
(3) Independent of the distance between the plates
(4) Directly proportional to potential difference between two plates

9. If separation between two plates of isolated charged parallel plate capacitor is doubled, then

- (1) The electrostatic force between two plates will be 4 times
(2) The potential difference between two plate will be doubled.
(3) Charge on the plates will be doubled.
(4) Electric field between two plates will be doubled

10. Three capacitors each of capacitance C were to be connected in such a way that

the effective capacitance becomes $\frac{3C}{2}$.

This can be done by:-

- (1) Connecting all of them in series
(2) Connecting them in parallel
(3) Connecting two in series and one in parallel
(4) Connecting two in parallel and one in series

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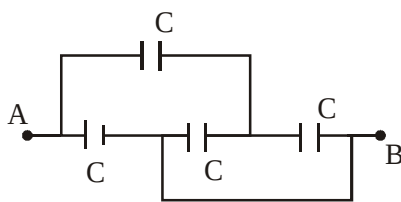
11. Twenty four capacitors each of capacitance $2\mu\text{F}$ are to be connected in such a way that the effective capacitance of $3\mu\text{F}$. This can be done by :-

- (1) Connecting 6 in series (in each rows) and 4 rows in parallel.
- (2) Connecting 8 in series (in each row) and 3 rows in parallel
- (3) Connecting 3 in series (in each row) and 8 rows in parallel
- (4) Connecting 4 in series (in each row) and 6 rows in parallel

12. A parallel plate air capacitor is charged by a battery. After disconnecting the battery, the distance between the plates of the capacitor is increased using an insulating handle. Choose the incorrect statement about this

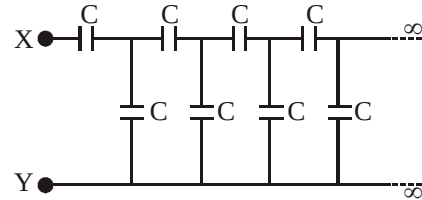
- (1) Electric field between plates remain unchanged
- (2) Potential difference between plates increases
- (3) Electrostatic force between plates decreases.
- (4) Electrostatic energy density remain unchanged.

13. The effective capacity of the network between terminals A and B in



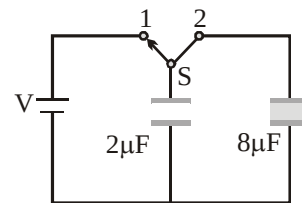
- (1) C (2) $\frac{3C}{2}$ (3) $\frac{5C}{3}$ (4) $\frac{3C}{5}$

14. In the adjoining figure, the effective capacity of the group of condensers will be :-



- (1) $2C$
- (2) ∞
- (3) zero
- (4) $0.62 C$

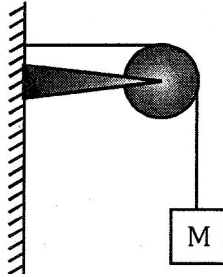
15. A $2\mu\text{F}$ capacitor is charged as shown in figure. The percentage of its stored energy dissipated after the switch S is turned to position 2 is:-



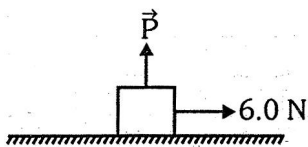
- (1) 0%
- (2) 20%
- (3) 75%
- (4) 80%

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16. A string of negligible mass going over a clamped pulley of mass m supports a block of mass M as shown in the figure. The force on the pulley by the clamp is given by :-



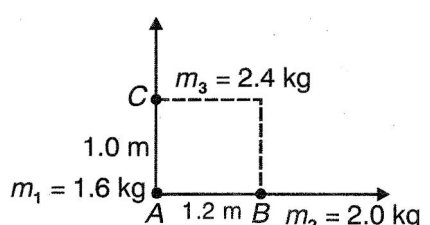
- (1) $\sqrt{2} Mg$
(2) $\sqrt{2} mg$
(3) $\left(\sqrt{(M+m)^2 + m^2}\right)g$
(4) $\left(\sqrt{(M+m)^2 + M^2}\right)g$
17. A 2.5 kg block is initially at rest on a horizontal surface. A 6.0 N horizontal force and a vertical force $\vec{P}$ are applied to the block as shown in figure. The coefficient of friction for the block and surface is 0.4. The magnitude of friction force when $P = 9\text{ N}$ ($g = 10\text{ m/s}^2$)



- (1) 6.0 N (2) 6.4 N
(3) 9.0 N (4) zero

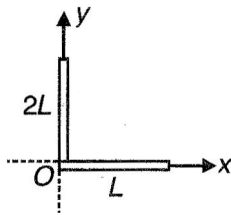
18. If K.E. increases by 4%. Then momentum will increase by :-
(1) 1.5% (2) 9%
(3) 3% (4) 2%
19. A force $\vec{F} = (3x^2 + 2x - 7)\text{ N}$ acts on a 2 kg body as a result of which the body gets displaced from $x = 0$ to $x = 5\text{ m}$. The work done by the force will be-
(1) 35 J (2) 70 J
(3) 115 J (4) 270 J
20. A string breaks if its tension exceeds 10 newtons. A stone of mass 250 gm tied to this string of length 10 cm is rotated in a horizontal circle. The maximum angular velocity of rotation can be :-
(1) 20 rad/s
(2) 40 rad/s
(3) 100 rad/s
(4) 200 rad/s
21. What is the value of linear velocity, if $\vec{\omega} = 3\hat{i} - 4\hat{j} + \hat{k}$ and $\vec{r} = 5\hat{i} - 6\hat{j} + 6\hat{k}$
(1) $6\hat{i} + 2\hat{j} - 3\hat{k}$
(2) $-18\hat{i} - 13\hat{j} + 2\hat{k}$
(3) $4\hat{i} - 13\hat{j} + 6\hat{k}$
(4) $6\hat{i} - 2\hat{j} + 8\hat{k}$
22. A force of $2\hat{i} + 3\hat{j} + 4\hat{k}\text{ N}$ acts on a body for 4 second, produces a displacement of $(3\hat{i} + 4\hat{j} + 5\hat{k})\text{ m}$. The power used is
(1) 9.5 W (2) 7.5 W
(3) 6.5 W (4) 4.5 W

SPACE FOR ROUGH WORK

23. Two equal masses m_1 and m_2 moving along the same straight line with velocities $+3 \text{ m/s}$ and -5 m/s respectively collide elastically. Their velocities after the collision will be respectively
- $+4 \text{ m/s}$ for both
 - -3 m/s and $+5 \text{ m/s}$
 - -4 m/s and $+4 \text{ m/s}$
 - -5 m/s and $+3 \text{ m/s}$
24. A body falling from a height of 10 m rebounds from hard floor. If it loses 20% energy in the impact, then coefficient of restitution is
- 0.89
 - 0.56
 - 0.23
 - 0.18
25. A tennis ball dropped from a height of 2 m rebounds only 1.5 m after hitting the ground. What fraction of its energy is lost in the impact
- $\frac{1}{4}$
 - $\frac{1}{2}$
 - $\frac{1}{3}$
 - $\frac{1}{8}$
26. A body of mass m moving with velocity v makes a head-on collision with another body of mass $2m$ which is initially at rest. The loss of kinetic energy of the colliding body (mass m) is
- $\frac{1}{2}$ of its initial kinetic energy
 - $\frac{1}{9}$ of its initial kinetic energy
 - $\frac{8}{9}$ of its initial kinetic energy
 - $\frac{1}{4}$ of its initial kinetic energy
27. Three point masses m_1 , m_2 and m_3 are placed at the corners of a thin massless rectangular sheet ($1.2 \text{ m} \times 1.0 \text{ m}$) as shown. Centre of mass will be located at the point
- 
- $(0.8, 0.6) \text{ m}$
 - $(0.6, 0.8) \text{ m}$
 - $(0.4, 0.4) \text{ m}$
 - $(0.5, 0.6) \text{ m}$

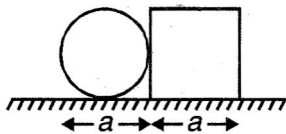
SPACE FOR ROUGH WORK

28. Figure shows a composite system of two uniform rods of lengths as indicated. Then the coordinates of the centre of mass of the system of rods are

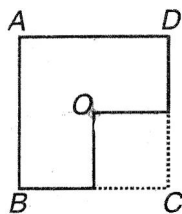


- (1) $\left(\frac{L}{2}, \frac{2L}{3}\right)$ (2) $\left(\frac{L}{4}, \frac{2L}{3}\right)$
(3) $\left(\frac{L}{6}, \frac{2L}{3}\right)$ (4) $\left(\frac{L}{6}, \frac{L}{3}\right)$

29. A circular plate of diameter 'a' is kept in contact with a square plate of side 'a' as shown. The density of the material and the thickness are same everywhere. The centre of mass of composite system will be



- (1) Inside the circular plate
(2) Inside the square plate
(3) At the point of contact
(4) Outside the system
30. From a uniform square plate, one-fourth part is removed as shown. The centre of mass of remaining part will lie on



- (1) OC (2) OA (3) OB (4) OD

31. Two particles A and B initially at rest move towards each other under a mutual force of attraction. At the instant when velocity of A is v and that of B is $2v$, the velocity of centre of mass of the system
- (1) v (2) $2v$ (3) $3v$ (4) Zero

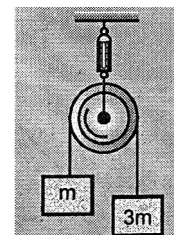
32. A ball of mass m is thrown upward and another ball of same mass is thrown downward so as to move freely under gravity. The acceleration of centre of mass is

- (1) g (2) $\frac{g}{2}$ (3) $2g$ (4) 0

33. The centre of mass of a system of three particles of masses $1g$, $2g$ and $3g$ is taken as the origin of a coordinate system. The position vector of a fourth particle of mass $4g$ such that the centre of mass of the four particle system lies at the point $(1, 2, 3)$ is $\alpha(\hat{i} + 2\hat{j} + 3\hat{k})$, where α is a constant. The value of α is :-

- (1) $\frac{10}{3}$ (2) $\frac{5}{2}$ (3) $\frac{1}{2}$ (4) $\frac{2}{5}$

34. If the system is released, then the acceleration of the centre of mass of the system :-



- (1) $\frac{g}{4}$ (2) $\frac{g}{2}$ (3) g (4) $2g$

SPACE FOR ROUGH WORK

35. A heavy nucleus at rest breaks into two fragments which fly off with velocities 8 : 1. The ratio of radii of the fragments is :-

(1) 1 : 2 (2) 1 : 4
(3) 4 : 1 (4) 2 : 1

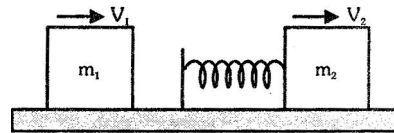
36. A sphere of mass m moving with a constant velocity collides with another stationary sphere of same mass. The ratio of velocities of two spheres after collision will be, if the co-efficient of restitution is e -

(1) $\frac{1-e}{1+e}$ (2) $\frac{e-1}{e+1}$
(3) $\frac{1+e}{1-e}$ (4) $\frac{e+1}{e-1}$

37. A particle falls from a height 'h' upon a fixed horizontal plane and rebounds. If V is the coefficient of restitution the total distance travelled before rebounding has stopped is :-

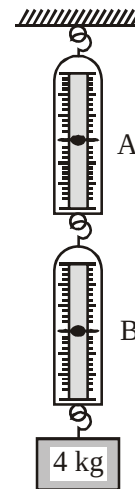
(1) $h \left(\frac{1+e^2}{1-e^2} \right)$ (2) $h \left(\frac{1-e^2}{1+e^2} \right)$
(3) $\frac{h}{2} \left(\frac{1-e^2}{1+e^2} \right)$ (4) $\frac{h}{2} \left(\frac{1+e^2}{1-e^2} \right)$

38. Two masses $m_1 = 2\text{kg}$ and $m_2 = 5\text{kg}$ are moving on a frictionless surface with velocities 10 m/s and 3 m/s respectively. m_2 is ahead of m_1 . An ideal spring of spring constant $k = 1120 \text{ N/m}$ is attached on the back side of m_2 . The maximum compression of the spring will be :-



(1) 0.51 m
(2) 0.062 m
(3) 0.25 m
(4) 0.72 m

39. A block of mass 4 kg is suspended through two light spring balances A and B. Then A and B will read respectively :-



(1) 4 kg and zero kg
(2) zero kg and 4 kg
(3) 4 kg and 4 kg
(4) 2 kg and 2 kg

40. One quantum of charge should be at least be equal to the charge in coulomb.

(1) $1.6 \times 10^{-17} \text{ C}$
(2) $1.6 \times 10^{-19} \text{ C}$
(3) $1.6 \times 10^{-10} \text{ C}$
(4) $4.8 \times 10^{-10} \text{ C}$

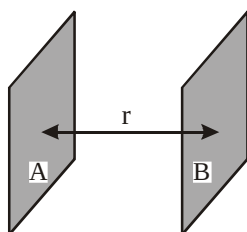
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41. A charged particle of charge q and mass m is released from rest in an uniform electric field E . Neglecting the effect of gravity, the kinetic energy of the charged particle after time ' t ' seconds is.

(1) $\frac{Eqm}{t}$ (2) $\frac{E^2 q^2 t^2}{2m}$

(3) $\frac{2E^2 t^2}{mq}$ (4) $\frac{Eq^2 m}{2t^2}$

42. The figure shows two parallel equipotential surfaces A and B kept a small distance r apart from each other. A point charge of q coulomb is taken from the surface A to B. The amount of net work done will be :



(1) $-\frac{1}{4\pi\epsilon_0} \frac{q}{r}$ (2) $\frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$

(3) $-\frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$ (4) zero

43. A particle of mass m and charge q is placed at rest in a uniform electric field E and then released. The KE attained by the particle after moving a distance y is :

(1) qEy^2 (2) qE^2y
(3) qEy (4) q^2Ey

44. Statement-1 : If Kinetic energy of a system is zero then linear momentum of each of particle of the system must be zero.
Statement-2 : If linear momentum of a system is zero then kinetic energy of system must be zero.

Statement-3 : If linear momentum of a particle is zero then kinetic energy of particle is zero.

Select the correct sequence.

- (1) TTF
(2) TFT
(3) FFT
(4) TFF

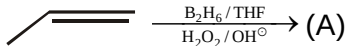
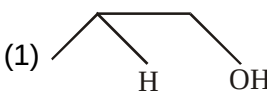
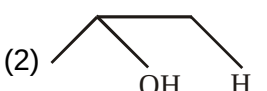
45. A particle of mass m moving with velocity u_1 collides elastically with another particle of same mass moving with velocity u_2 in the same direction. After collision their speeds are v_1 and v_2 respectively then

(A) $u_1 + v_1 = v_2 + u_2$

(B) $u_1 - v_1 = v_2 + u_2$

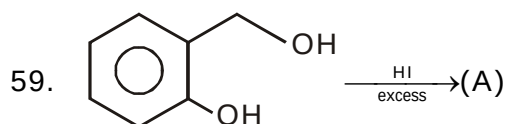
- (1) Both the equation A and B are correct
(2) Both the equations A and B are incorrect
(3) Equation A is correct but not B
(4) Equation B is correct but not A

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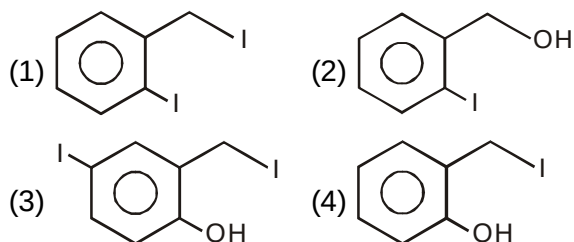
46. Mass of 1 atom of Hydrogen is -
 (1) 1.66×10^{-24} g (2) 10^{-22} g
 (3) 10^{-23} g (4) 10^{-25} g
47. What is correct for 10 g of CaCO_3 -
 (1) It contains 1g-atom of carbon
 (2) It contains 0.3 g-atoms of oxygen
 (3) It contains 12 g of calcium
 (4) None of these
48. Equal masses of O_2 , H_2 and CH_4 are taken in a container. The respective mole ratio of these gases in container is -
 (1) 1 : 16 : 2 (2) 16 : 1 : 2
 (3) 1 : 2 : 16 (4) 16 : 2 : 1
49. The mass percent of N in 66% pure $(\text{NH}_4)_2\text{SO}_4$ sample is -
 (1) 32 (2) 28
 (3) 14 (4) None of these
50. The law of conservation of mass holds good for all of the following except -
 (1) All chemical reactions
 (2) Nuclear reactions
 (3) Endothermic reactions.
 (4) Exothermic reactions.
51. If LPG cylinder contains mixture of butane and isobutane, then the amount of oxygen that would be required for combustion of 1kg of it will be-
 (1) 1.8 kg (2) 2.7 kg
 (3) 4.5 kg (4) 3.58 kg
52. 8 gm of O_2 has the same number of molecules as-
 (1) 7 gm of CO (2) 14 gm of CO
 (3) 14 gm of CO_2 (4) 12 gm of CO_2
53. A sample of hard water is found to contain 40 mg of Ca^{+2} ion per litre. The amount of washing soda (Na_2CO_3) required to soften five litres of the sample would be -
 (1) 1.06 gm (2) 5.3 gm
 (3) 53 mg (4) 530 mg
54. 1.5 gm of divalent metal displaced 4 gm of copper (at. wt. = 64) from a solution of copper sulphate. The atomic weight of the metal is-
 (1) 12 (2) 24 (3) 48 (4) 6
55. Avogadro's number of Rupees can be spent inyears if 10 lac rupees per second are spent -
 (1) 1.91×10^{10} year
 (2) 2.91×10^{10} year
 (3) 3.91×10^{10} year
 (4) 4.91×10^{10} year
56.  $\xrightarrow[\text{H}_2\text{O}_2/\text{OH}^-]{\text{B}_2\text{H}_6/\text{THF}}$ (A)
 Product 'A' is :-
 (1) 
 (2) 
 (3) Both (1) & (2)
 (4) Ethene
57. Secondary alcohol on victor mayer test gives :-
 (1) Red colour
 (2) Colourless solution
 (3) Blue colour
 (4) Pink colour

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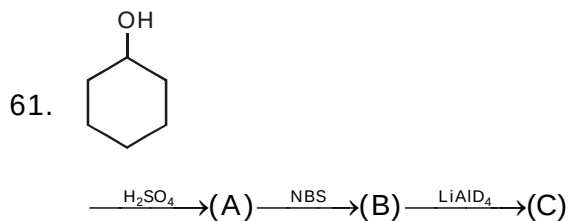
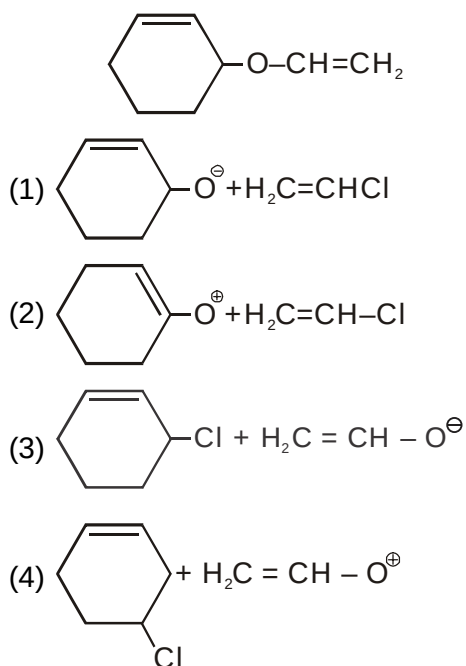
58. OsO_4 is used for
 (1) Hydroxylation of acetylenes
 (2) Hydroxylation of olefins to form cis-diol
 (3) Hydroxylation of olefine to form trans-diol
 (4) Hydroxylation of carbonyl compounds



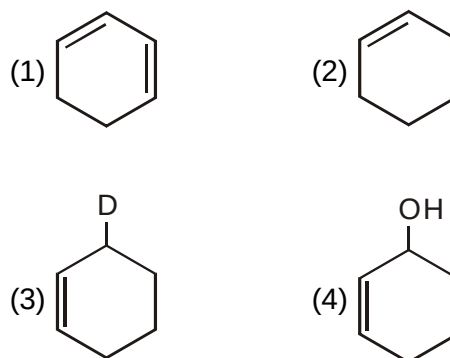
Major Product (1) is :-



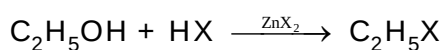
60. The best method for synthesis of given ether by Williamson's ether sythesis is :-



Product (3) is :-



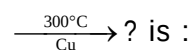
62. For the reaction,



The decreasing order of reactivity of halogen acids is

- (1) $\text{HI} > \text{HCl} > \text{HBr}$
 (2) $\text{HI} > \text{HBr} > \text{HCl}$
 (3) $\text{HCl} > \text{HBr} > \text{HI}$
 (4) $\text{HBr} > \text{HI} > \text{HCl}$

63. The product of reaction $\text{CH}_3\text{CH}_2\text{OH}$



- (1) C_2H_6
 (2) CH_3COCH_3
 (3) CH_3COOH
 (4) CH_3CHO

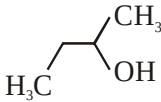
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64. In the following sequence of reactions,
 $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH} \xrightarrow{\text{P+I}_2} \text{A} \xrightarrow[\text{ether}]{\text{Mg}} \text{B}$
 $\xrightarrow{\text{HCHO}} \text{C} \xrightarrow{\text{H}_2\text{O}} \text{D}$, then compound 'D' is

- (1) butanal
- (2) n-butyl alcohol
- (3) n-propyl alcohol
- (4) propanal

65. Which of the following is a tertiary alcohol ?

- (1) 2-Methylpropan-1-ol
- (2) 2-Methylpropan-2-ol
- (3) 3-Methylbutan-1-ol
- (4) 3-Methylbutan-2-ol

66.  $\xrightarrow[-\text{H}_2\text{O}]{\text{H}^+} [\text{F}] \xrightarrow[\text{CCl}_4]{\text{Br}_2} \text{C}_4\text{H}_8\text{Br}_2$
 5 such products are possible

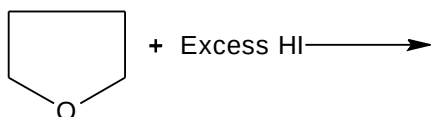
How many isomers of [F] are possible ?

- (1) 2
- (2) 5
- (3) 6
- (4) 3

67. The acidic character of 1° , 2° , 3° alcohols H_2O and $\text{RC} \equiv \text{CH}$ is in the order

- (1) $\text{H}_2\text{O} > 1^\circ > 2^\circ > 3^\circ > \text{RC} \equiv \text{CH}$
- (2) $\text{RC} \equiv \text{CH} > 3^\circ > 2^\circ > 1^\circ > \text{H}_2\text{O}$
- (3) $1^\circ > 2^\circ > 3^\circ > \text{H}_2\text{O} > \text{RC} \equiv \text{CH}$
- (4) $3^\circ > 2^\circ > 1^\circ > \text{H}_2\text{O} > \text{RC} \equiv \text{CH}$

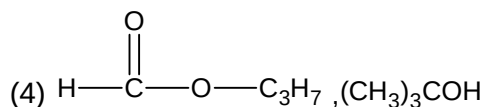
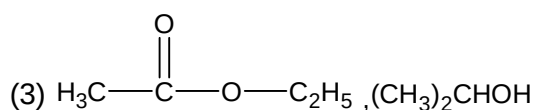
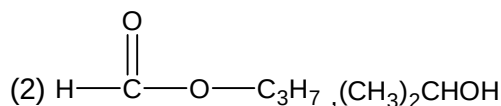
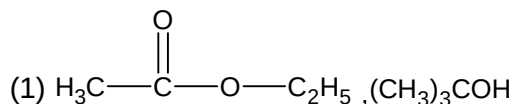
68. Predict the major product



- (1) $\text{HO}-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{I}$
- (2) $\text{HO}-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{OH}$
- (3) $\text{I}-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{I}$
- (4) no reaction

69. Ester (A) $\text{C}_4\text{H}_8\text{O}_2 + \text{CH}_3\text{MgBr} \xrightarrow[\text{(2 parts)}]{\text{H}_3\text{O}^+} \text{(B)} \text{C}_4\text{H}_{10}\text{O}$

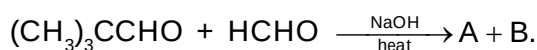
Alcohol B reacts fastest with Lucas reagent. Hence A and B are



70. Acetic acid when heated (300°C) with MnO gives

- (1) formaldehyde
- (2) acetaldehyde
- (3) acetone
- (4) butaone

71. In the reaction,

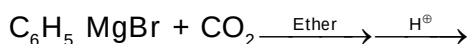


the products (A) and (B) are respectively

- (1) $(\text{CH}_3)_3\text{CCH}_2\text{OH}$ and $\text{HCOO}^- \text{Na}^+$.
- (2) $(\text{CH}_3)_3\text{CCOONa}$ and CH_3OH .
- (3) $(\text{CH}_3)_3\text{CCH}_2\text{OH}$ and CH_3OH .
- (4) $(\text{CH}_3)_3\text{COONa}$ and $\text{HCOO}^- \text{Na}^+$.

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72. In the following reaction final product is :



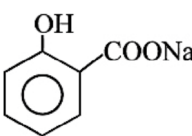
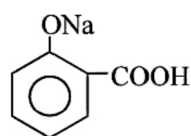
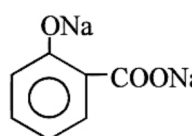
Find product

- (1) Benzoic acid
- (2) Benzaldehyde
- (3) Benzamide
- (4) Benzene

73. In which of the following reaction the final product is neither an acid nor an acid salt.

- (1) $\text{Ph}-\text{CHO} \xrightarrow{\text{Tollen's reagent}}$
- (2) $\text{CH}_3-\text{CH}_2-\text{OH} \xrightarrow{\text{KMnO}_4/\text{OH}^-}$
- (3) $\text{Ph}-\text{CHO} \xrightarrow{\text{Fehling solution}}$
- (4) $\text{Ph}-\text{CH}_2-\text{OH} \xrightarrow{\text{K}_2\text{Cr}_2\text{O}_7/\text{H}^+}$

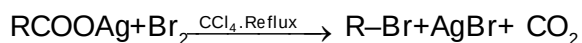
74. Sodium bicarbonate reacts with salicylic acid to form:

- (1) $\text{C}_6\text{H}_5\text{ONa}$
- (2) 
- (3) 
- (4) 

75. Which of the following will not undergo Hell-Volhard Zelinsky (HVZ) reaction ?

- (1) HCOOH
- (2) CH_3COOH
- (3) $\text{CH}_3\text{CH}_2\text{COOH}$
- (4) $\text{CH}_3\text{CHBrCOOH}$

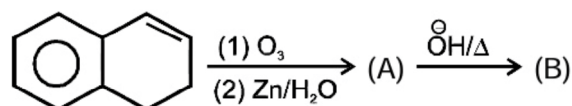
76. The reaction :

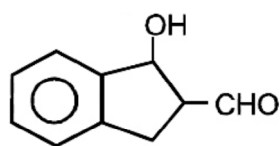
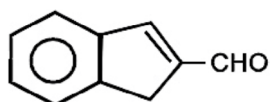
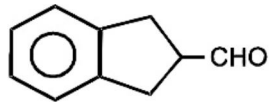
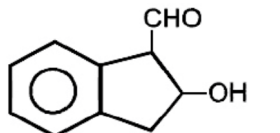


is called

- (1) Wurtz reaction
- (2) Hunsdiecker bromo decarboxylation reaction
- (3) Friedel-Crafts reaction
- (4) Kolbe's reaction

77. In the given reaction sequence B is



- (1) 
- (2) 
- (3) 
- (4) 

78. Hoffmann bromamide degradation reaction is shown by _____.

- (1) ArNH_2
- (2) ArCONH_2
- (3) ArNO_2
- (4) ArCH_2NH_2

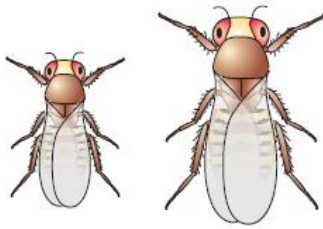
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79. The compound that can work both as an oxidizing agent and a reducing agent is
 (1) KMnO_4 (2) H_2S
 (3) BaO_2 (4) H_2O_2
80. The number of moles of KMnO_4 that will be needed to react completely with one mole of ferrous oxalate in acidic medium is
 (1) $3/5$ (2) $2/5$
 (3) $4/5$ (4) 1
81. For the reaction
 $\text{MnO}_4^- + \text{C}_2\text{O}_4^{2-} + \text{H}^+ \longrightarrow \text{Mn}^{2+} + \text{CO}_2 + \text{H}_2\text{O}$
 The correct coefficients of the reactants for the balanced reaction are

MnO_4^-	$\text{C}_2\text{O}_4^{2-}$	H^+
(1) 2	5	16
(2) 16	5	2
(3) 5	16	2
(4) 2	16	5
82. Oxidation states of carbon atoms in diamond and graphite are :-
 (1) +2, +4 (2) +4, +2
 (3) -4, 4 (4) zero, zero
83. The oxidation number of Cr in CrO_5 is
 (1) +3 (2) +5 (3) +6 (4) 0
84. In the reaction,
 $3\text{Cl}_2 + 6\text{NaOH} \longrightarrow \text{NaClO}_3 + 5\text{NaCl} + 3\text{H}_2\text{O}$
 the element which loses as well as gains electrons is
 (1) Na (2) O
 (3) Cl (4) none of three
85. In which one of the following compounds the oxidation number of oxygen is positive?
 (1) H_2O_2 (2) Na_2O_2
 (3) OF_2 (4) H_2O
86. Equivalent weight of Mn^{3+} in the following reaction is ($\text{Mn} = 55$)
 $\text{Mn}^{3+} \longrightarrow \text{Mn}^{2+} + \text{MnO}_2$
 (1) 27.5 (2) 55
 (3) 110 (4) 165
87. The reaction $\text{H}_2\text{S} + \text{H}_2\text{O}_2 \longrightarrow 2\text{H}_2\text{O} + \text{S}$ shows :-
 (1) oxidising nature of H_2O_2
 (2) reducing action of H_2O_2
 (3) alkaline nature of H_2O_2
 (4) acidic nature of H_2O_2
88. Oxidation numbers of two Cl atoms in bleaching powder, CaOCl_2 is -
 (1) -1, -1
 (2) +1, -1
 (3) +1, +1
 (4) 0, -1
89. Oxygen shows oxidation state of -1 in the compound -
 (1) NO_2
 (2) MnO_2
 (3) PbO_2
 (4) Na_2O_2
90. Equivalent weight of FeC_2O_4 in the change,
 $\text{FeC}_2\text{O}_4 \longrightarrow \text{Fe}^{3+} + \text{CO}_2$ is
 (1) $M/3$ (2) $M/6$
 (3) $M/2$ (4) $M/1$

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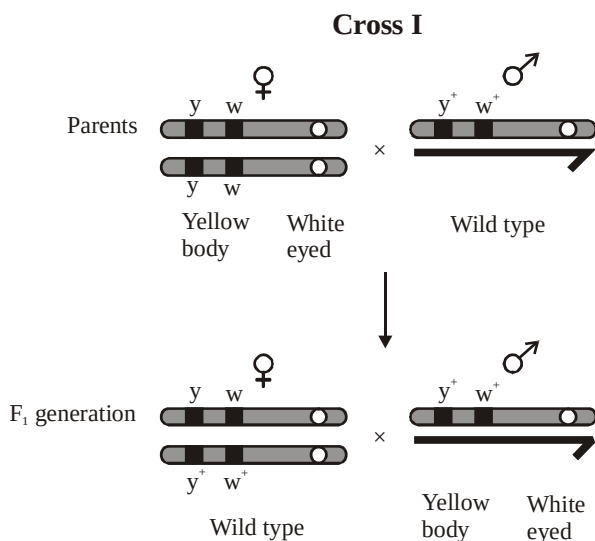
91. Given below is, the figure of *Drosophila melanogaster*.



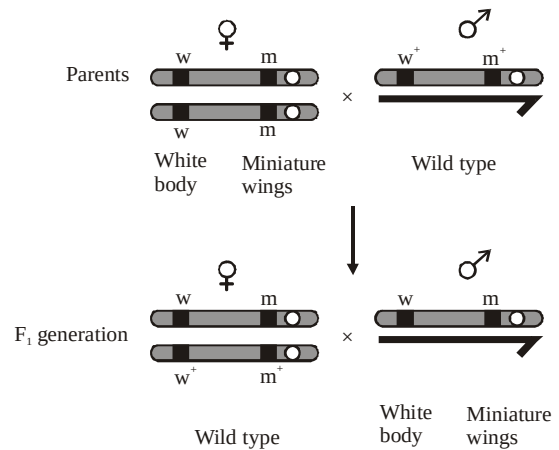
It was found to be very suitable for experimental verification of chromosomal theory of inheritance by Morgan, because.

- (1) A single mating produces two young flies
- (2) Smaller female is easily recognisable from larger male
- (3) It completes life cycle in about two weeks
- (4) It reproduces parthenogenetically

92. The experiment shown in the figure has been carried out by Morgan to show the phenomenon of linkage & recombination. If in Cross I, genes are tightly linked and in Cross II, genes are loosely linked then what will be percentage of recombinants produced in Cross I and Cross II respectively in F_2 gene ration ?

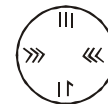


Cross II



- (1) 98.7% and 62.8% (2) 1.3% and 37.2%
- (3) 37.2 and 1.3% (4) 62.8% and 98.7%

93. Given below is the karyotype of *Drosophila*.



The sex of such *Drosophila* should be –

- (1) Superfemale
 - (2) Intersex
 - (3) Sterile male
 - (4) Supermale
94. Crossing over is advantageous because it brings about
- (1) variation
 - (2) linkage
 - (3) inbreeding
 - (4) stability
95. Two genes a and b are linked and show 30% recombination. If ++ / ++ individual married with ab / ab then the types and proportion of gametes in F_1 will be -
- (1) ++ 35% : ab : 35% : + a 15% : +b 15%
 - (2) ++ 15% : ab : 15% : + a 35% : +b 35%
 - (3) ++ 35% : ab : 15% : + a 35% : +b 15%
 - (4) ++ 15% : ab : 35% : + a 15% : +b 35%
96. Coupling and Repulsion theory produced by:-

- (1) Morgan (2) Bateson
- (3) Muller (4) De vries

97. Which statement is not true for *Drosophila melanogaster*–

- (1) They complete their life cycle about two weeks
- (2) Single mating produce large number of progeny flies
- (3) It has few hereditary variation that can be seen with high power microscope
- (4) It has clear differentiation of the sex

98. In *Opuntia*, the function of photosynthesis is carried out by

- (1) Cladode
- (2) Phylloclade
- (3) Phyllode
- (4) Bulb

99. Match the following and select the correct combination from the options given below

Column I (Stem Modifications)	Column II (Found in)
A. Underground stem	1. Euphorbia
B. Stem tendril	2. Opuntia
C. Stem thorns	3. Potato
D. Flattened stem	4. Citrus
E. Fleshy cylindrical stem	5. Cucumber

- (1) A-1, B-2, C-3, D-5, E-4
- (2) A-2, B-3, C-4, D-5, E-1
- (3) A-3, B-4, C-5, D-1, E-2
- (4) A-3, B-5, C-4, D-2, E-1

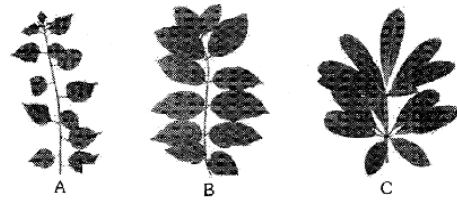
100. Ginger is a stem which can be differentiated from root because it

- (1) Grows parallel to ground
- (2) Stores food
- (3) Lacks chlorophyll
- (4) Has nodes and internodes

101. Find the correct match

- (1) Mustard plant: leaves are opposite
- (2) Mustard plant: leaves are alternate
- (3) Guava plant: Leaves are alternate
- (4) Guava plant: Leaves are whorled

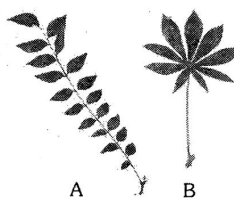
102. Identify the correct types of phyllotaxy which shown in the following figures



- (1) A - Whorled, B - Alternate, C - Opposite
- (2) A - Alternate, B - Whorled, C - Opposite
- (3) A - Whorled, B - Opposite, C - Alternate
- (4) A - Alternate, B - Opposite, C – Whorled

103. In sweet pea, the tendrils are modified

- (1) Stem branches
- (2) Leaflets
- (3) Leaves
- (4) Stipules



104.

See the following figures and identify leaves A and B

- (1) A - Palmately compound leaf
B - Palmately compound leaf
- (2) A - Pinnately compound leaf,
B - Pinnately compound leaf
- (3) A - Palmately compound leaf,
B - Pinnately compound leaf
- (4) A - Pinnately compound leaf,
B - Palmately compound leaf

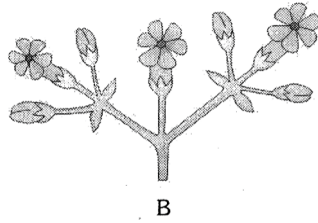
105. In cyathium inflorescence

- (1) Single male flower is surrounded by female flowers
- (2) Male and female flowers are borne in different plants
- (3) There is of one male and one female flowers
- (4) Single female flower surrounded by many peripheral male flowers

106. See the following diagrams and identify inflorescence A and B



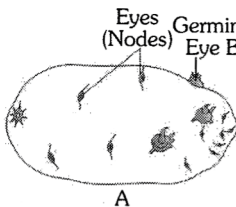
A



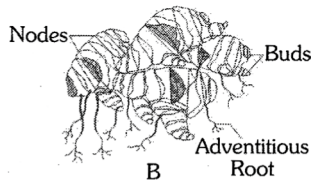
B

- (1) A - Cymose, B - Cymose
- (2) A - Racemose, B - Racemose
- (3) A - Racemose, B - Cymose
- (4) A - Cymose, B - Racemose

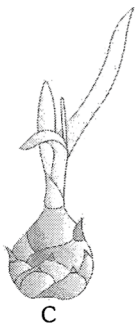
107. Study the diagrams given below and select the right options out of (A – D); in which all the 4 items A, B, C and D are correctly identified



A



B



C

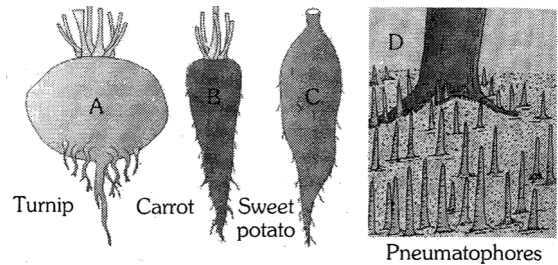


D

A B C D

- (1) Offset Sucker Stolon Leaf buds
- (2) Offset Sucker Stolon Leaf buds
- (3) Tuber Rhizome Bulb Leaf buds
- (4) Tuber Rhizome Bulbil Leaf buds

108. See the following diagrams Which of the following is not correct about A, B, C and D



- (1) A, B and C are underground roots but D grows vertically upwards
- (2) Pneumatophore is found in the plants that grow in sandy soil
- (3) Pneumatophores help to get oxygen for respiration
- (4) Tap roots of carrot, turnip and adventitious root of sweet potato, get swollen and store food

109. Which of the following is not a stem modification

- (1) Pitcher of Nepenthes
- (2) Thorns of citrus
- (3) Tendrils of cucumber
- (4) Flattened structures of Opuntia

110. Rachis is present in

- (1) Pinnate compound leaf
- (2) Palmate compound leaf
- (3) Both correct
- (4) Both wrong

111. Onion stores food in

- (1) Underground stem
- (2) Fleshy scales
- (3) Root
- (4) Shoot

112. This is an example of group of plants possessing stilt root.

- (1) Orchid and pothos
- (2) Maize and pandanus
- (3) Banyan and Tinospora
- (4) Cascuta and Ioranthus

113. What is meant by condensed form of rhizome growing vertically.

- (1) Rhizom (2) Tuber
(3) Corm (4) Bulbil

114. In Given following stem and their example select right option with suitable example

- (1) Runner – Potato, Zamikand
(2) Sucker – Colocasia, Onion
(3) Bulb – Turmeric, Ginger
(4) Offset – Pistia, Eichhornia

115. Winged petiole found in

- (1) China rose (2) Citrus
(3) Eichhornia (4) Ginger



116.

Part A is a modified.

- (1) leaf let (2) leaf
(3) stem (4) leaf base



117.

Part A is a modification of

- (1) leaf let hooks (2) Hooks (stem)
(3) leaf spines (4) Prickles (stem)

118.



Above given figure is modification of stem.
The purpose of this modification is

- (1) Storage
(2) Support climbing
(3) Protection, Climbing
(4) Spread & vegetative propagation

119. Reverse transcriptase is

- (1) RNA dependent RNA polymerase
(2) DNA dependent RNA polymerase
(3) DNA dependent DNA polymerase
(4) RNA dependent DNA polymerase

120. Initiation of polypeptide chain in protein synthesis is induced by

- (1) methionine (2) glycine
(3) leucine (4) lysine

121. Khorana synthesized a biologically functional tyrosine suppressor t-RNA of *E. coli* in 1979 it contained.

- (1) 77 nucleotide pairs
(2) 207 nucleotide pairs
(3) 312 nucleotide pairs
(4) 333 nucleotides only

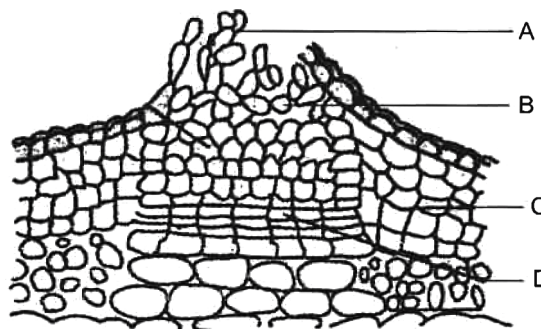
122. From a DNA template with the sequence CTGATAGC, the mRNA sequence formed would be

- (1) GUCTUTCG (2) GACUAUCG
(3) UACTATCU (4) GAUTATUG

123. Which is the correct sequence of code transfer involved in the formation of a polypeptide ?

- (1) DNA, tRNA, tRNA, mRNA
(2) rRNA, DNA, mRNA, tRNA
(3) mRNA, tRNA, DNA, amino acids
(4) DNA, mRNA, tRNA, amino acids

124. rRNA except 5 s rRNA in eukaryotes is synthesized by
 (1) RNA polymerase I
 (2) RNA polymerase II
 (3) RNA polymerase III
 (4) DNA polymerase
125. In DNA Replication RNA primer is synthesized by
 (1) DNA polymerase I
 (2) DNA polymerase III
 (3) primosome
 (4) primase
126. The coding sequences of mRNA is
 (1) intron (2) exon (3) muton (4) cistron
127. The removal of non-coding sequence from primary eukaryotic RNA sequence is called
 (1) splicing
 (2) post transcriptional modification
 (3) intranuclear traffic control signal
 (4) none of the above
128. Interfascicular cambium develops from the cells of :-
 (1) Pericycle
 (2) Medullary rays
 (3) Xylem parenchyma
 (4) Endodermis
129. Select the correct statement from the following
 (1) All lateral meristems are secondary meristem
 (2) All secondary meristems are lateral meristem
 (3) All intercalary meristems are secondary meristem
 (4) All secondary meristems are intercalary meristem.
130. Which of the following tissues are included in bark? (According to modern view)
 (1) Secondary xylem, periderm and secondary phloem
 (2) Secondary cortex, secondary phloem and xylem.
 (3) Dermatogen, periblem, plerome and secondary phloem.
 (4) Phellem, Phellogen, Phelloderm and secondary phloem.
131. Which of the following are examples of lateral meristems
 (1) Fascicular vascular combium
 (2) Inter-fascicular cambium
 (3) Cork cambium
 (4) All of the above
132. The autumn wood is A in colour and has a B density. A and B are respectively :-
 (1) Lighter, Higher
 (2) Darker, Higher
 (3) Lighter, Lower
 (4) Darker, Lower
133. The portion of the oldest layer of secondary xylem is :
 (1) Just inside the vascular cambium
 (2) Just inside the primary xylem
 (3) Just outside the vascular cambium
 (4) Just outside the primary xylem
134. All the tissues exterior to vascular cambium are termed as
 (1) Phelloderm
 (2) Periderm
 (3) Bark
 (4) Cork
135. In the given diagram, identify A, B, C and D.



	A	B	C	D
(1)	Epidermis	Complementary cells	Phellogen	Phelloderm
(2)	Epidermis	Phellem	Phellogen	Phelloderm
(3)	Phelloderm	Periderm	Phellogen	Phellem
(4)	Epidermis	Cork	Cork cambium	Secondary cortex

136. Which of the following is wrong statement about hyoid bone ?

- (1) Only one in number
- (2) U-shaped bone
- (3) Present at the top of the buccal cavity
- (4) Not attached with any other bone of the body

137. Our vertebral column is formed of :

- (1) 22 vertebra
- (2) 26 vertebra
- (3) 32 vertebra
- (4) 36 vertebra

138. Saddle joint occurs between :

- (1) Carpal and metacarpal of thumb
- (2) Femur and pelvic girdle
- (3) All the vertebrae
- (4) Phalanges

139. The correct formula for human vertebral column is :

- (1) $C_{14} T_{12} L_5 S_4 Coc_1$
- (2) $C_7 T_{14} L_5 S_1 Coc_1$
- (3) $C_7 T_{12} L_5 S_1 Coc_1$
- (4) $C_7 T_{14} L_5 S_4 Coc_1$

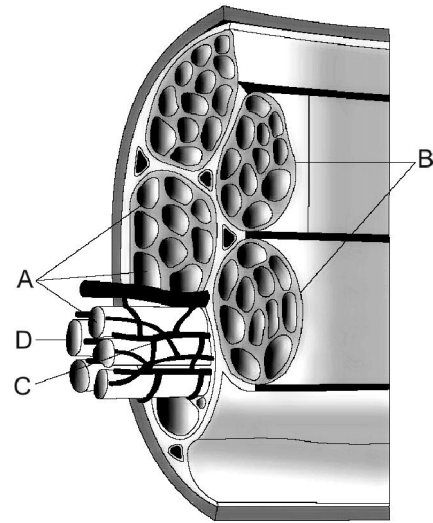
140. The appendicular skeleton includes :

- (1) The bones of the limbs only
- (2) The bones of the limbs along with their girdles
- (3) The skull, vertebral column, sternum and ribs
- (4) None of these

141. Select incorrect statement :

- (1) Axial skeleton comprises of 80 bones
- (2) Each middle ear contain three pairs of tiny bones - malleus, incus and stapes
- (3) The skull is composed of bones - cranial, facial, Hyoid and ear ossicles
- (4) Acromian process is present in scapula

142. Identify A, B, C & D :



- (1) A = Sarcolemma, B = Fascicle (muscle bundle), C = Blood capillary, D = Muscle fibre (muscle cell)
- (2) A = Muscle fibre (muscle cell), B = Fascicle (muscle bundle), C = Blood capillary, D = Sarcolemma
- (3) A = Fascicle (muscle bundle), B = Muscle fibre (muscle cell), C = Sarcolemma, D = Blood capillary
- (4) A = Sarcolemma, B = Muscle fibre (muscle cell), C = Blood capillary, D = Fascicle (muscle bundle)

143. The cells of human body exhibit :

- (1) Amoeboid movement
- (2) Ciliary movement
- (3) Muscular movement
- (4) All of these

144. Myosin head contains :

- (1) ATPase
- (2) ATP binding site
- (3) Both (1) & (2)
- (4) None of these

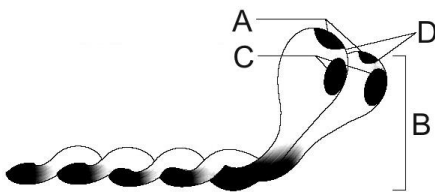
145. Locomotion requires a perfect co-ordinated activity of :

- (1) Muscular system
- (2) Skeletal system
- (3) Neural system
- (4) All of these

146. Which one of the following is the correct description of a certain part of a normal human skeleton :

- (1) First vertebrae is axis which articulates with the occipital condyles
- (2) The 9th and 10th pair of ribs are called the floating ribs
- (3) Glenoid cavity is a depression to which the thigh bone articulates
- (4) Parietal bone and the temporal bone of the skull are joined by fibrous joint

147. Identify the A, B, C & D in given diagram :



- (1) A = Actin binding site, B = Cross arm, C = ATP binding site, D = Head
- (2) A = Cross arm, B = Actin binding site, C = ATP binding site, D = Head
- (3) A = Head, B = Actin binding site, C = Cross arm, D = ATP binding site
- (4) A = ATP binding site, B = Head, C = Actin binding site, D = Cross arm

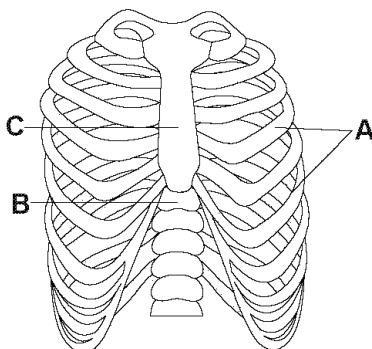
148. Which of the following is called collar bone?

- (1) Scapula
- (2) Atlas
- (3) Clavicle
- (4) Axis

149. Which cartilage covers the articulating surfaces of joint:

- (1) Elastic cartilage
- (2) Fibrous cartilage
- (3) Hyaline cartilage
- (4) Its covered by bones not from cartilage

150. Identify the A, B & C :



- (1) A = Sternum, B = Vertebral Column, C = Ribs
- (2) A = Vertebral Column, B = Sternum, C = Ribs
- (3) A = Ribs, B = Vertebral Column, C = Sternum
- (4) A = Sternum, B = Ribs, C = Vertebral Column

151. Vertebro-chondrial ribs are :

- (1) 8th, 9th & 10th pairs
- (2) 1st to 7th pairs
- (3) 11th & 12th pairs
- (4) All ribs are vertebro-chondrial ribs

152. Muscles are classified as red and white fibres based on:

- (1) Numbers of mitochondria
- (2) Fibres of muscles
- (3) Myoglobin pigment
- (4) All of these

153. In the centre of each band is an elastic fibre called line which bisect it :

- (1) I, Z
- (2) A, Z
- (3) I, M
- (4) More than two are correct

154. Each pectoral girdle is consist of :

- (1) Clavicle, Scapula, Ribs, Sternum
- (2) Scapula, Ribs, Cervical vertebra
- (3) Scapula and Clavicle only
- (4) Scapula and ribs only

155. Choose incorrect statements :

- (1) Troponin is distributed at regular intervals on the tropomyosin
- (2) Each vertebra has a central neural canal through which spinal cord passes
- (3) Axis bone articulate with occipital condyles
- (4) All are correct

156. Which points are not correct about Red fibres :-
 (i) More myoglobin
 (ii) More mitochondria
 (iii) More sarcoplasmic reticulum
 (iv) Depend on anaerobic process for energy
 (v) Gives a reddish appearance.
 (1) (i), (ii) (2) (iii), (iv)
 (3) (ii), (iii) (4) (iv), (v)
157. During muscle contraction, energy is provided by :-
 (1) Acetyl-Co.A
 (2) Phosphogen
 (3) AMP
 (4) Glucose
158. Muscular and nervous excitability is reduced by:-
 (1) Na^+ (2) K^+
 (3) Ca^{++} (4) Mg^{++}
159. Contractile protein of skeletal muscle involving ATPase activity is :-
 (1) Troponin (2) Tropomyosin
 (3) Myosin (4) α -actinin
160. One myosin filament in the myofibril of skeletal muscle fibre is surrounded by how many actin filaments ?
 (1) Two (2) Four
 (3) Six (4) Three
161. Ions used for muscle contraction are
 (1) Na^+ and Ca^{2+} (2) Ca^{2+} and Mg^{2+}
 (3) Mg^{2+} and Na^+ (4) Ca^{2+} and K^+
162. Regulation of kidney function of Juxtaglomerular Apparatus (JGA) involves certain steps given below. Arrange them in the correct order :-
 (a) Release of enzyme renin
 (b) Release of aldosterone from adrenal gland
 (c) Reabsorption of Na^+ and water at DCT
 (d) Decrease in blood pressure and blood volume
 (e) Conversion of angiotensinogen to angiotensin II
- (1) a, e, b, c, d
 (2) b, c, a, e, d
 (3) d, a, b, e, c
 (4) d, a, e, b, c
163. Which one of the following statements in regard to the excretion by the human kidney is correct?
 (1) Ascending limb of Loop of Henle is impermeable to electrolytes
 (2) Descending limb of Loop of Henle is impermeable to water
 (3) Distal convoluted tubule is incapable of reabsorbing HCO_3^-
 (4) Nearly 99 per cent of the glomerular filtrate is reabsorbed by the renal tubules
164. Internal section of Kidney shows extension of cortex into medullary regions between the pyramids and is called
 (1) Ducts of Bellini
 (2) Major calyx
 (3) Minor calyx
 (4) Columns of Bertini
165. The protective layer of Kidney is/are:
 (1) Renal capsule
 (2) Adipose tissue
 (3) Renal fascia
 (4) All of these
166. Which of the following epithelium is present in the inner lining of urinary bladder?
 (1) Columnar
 (2) Cuboidal
 (3) Transitional
 (4) Pseudostratified
167. The structural and functional unit of Kidney is called
 (1) Nephron
 (2) Uriniferous tubule
 (3) Both (1) and (2)
 (4) Malpighian tubule

168. Kidneys are vital organs of the body because they help in
 (1) Removal of metabolic wastes
 (2) Regulation of pH
 (3) Maintain level of body fluids
 (4) All of these
169. The renal threshold for glucose is
 (1) 100 mg/dL
 (2) 120 mg/dL
 (3) 180 mg/dL
 (4) 250 mg/dL
170. Accumulation of urea in blood of patients suffering with renal failure is called
 (1) Renal calculi
 (2) Glomerulonephritis
 (3) Uremia
 (4) Ketosis
171. Different types of excretory structures and animals are given below. Match them appropriately and mark the correct answer from among those given below:
 Excretory structure Animals
 A. protonephridia i. Prawn
 B. Nephridia ii. Cockroach
 C. Malpighian tabules iii. Earthworm
 D. Green gland iv. Flatworms
 (1) (D) i, (C) ii, (B) iii and (A) iv
 (2) (B) i, (C) ii, (A) iii and (B) iv
 (3) (D) i, (C) ii, (A) iii and (B) iv
 (4) (B) i, (C) ii, (B) iii and (D) iv
172. Which one of the following is also known as antidiuretic hormone?
 (1) Oxytocin
 (2) Vasopressin
 (3) Adrenaline
 (4) Calcitonin
173. Diaphragms, cervical caps and vaults prevent conception by :-
 (1) Increasing phagocytosis of sperms within uterus
 (2) Suppressing sperm motility
 (3) Inhibiting ovulation and implantation
 (4) Blocking the entry of sperms through the cervix
174. The 'family planning' programmes were periodically assessed over the past decades. Improved programmes covering wider reproductive-related areas are currently in operation under the popular name :-
 (1) RCH programmes
 (2) ART programmes
 (3) MTP programmes
 (4) Test tube baby programmes
175. Why the coitus interruptus method shows very high failure rate ?
 (1) Coitus interruptus is not possible
 (2) Initial secretion in male, may also contain sperms
 (3) Capacitation occurs in vagina
 (4) None of the above
176. Cu ions released from copper-releasing Intra Uterine Devices (IUDs) :-
 (1) Prevent ovulation
 (2) Make fallopian tube unsuitable for fertilization
 (3) Increase phagocytosis of sperms
 (4) Suppress sperm motility
177. What is the action of SAHELI ?
 (1) Antiestrogenic
 (2) Antiprogestronic
 (3) Releasing progesterone prevents implantation
 (4) Releasing copper inhibits motility of sperms

178. Consider the following statements each with one or two blanks.

- (i) Towards the centre of the inner concave surface of the kidney is a notch called(a)..... through which ureter, blood vessels and nerves enter.
- (ii) The medulla of kidney is divided into a few conical masses called(b)..... projecting into the(c)..... .
- (iii) Glomerulus is a tuft of capillaries formed by the(d)..... arteriole, Blood from the glomerulus is carried away by an(e).... arteriole.

Which one of the following options correctly fills the blanks in any two of the statements ?

- (1) (a) - renal pelvis, (b) - calyces, (c) medullary pyramids
- (2) (b) - medullary pyramids, (c) - calyces, (d) - afferent, (e) - efferent
- (3) (b) - columns of Bertini, (c) - chordae tendinde, (d) efferent, (e) - afferent
- (4) (a) - hilum, (d) - efferent, (e) - afferent

179. Which one of the following statements is correct with respect to kidney function regulation?

- (1) When someone drinks lot of water, ADH releasees increased.
- (2) Exposure to cold temperature stimulates ADH release.
- (3) A decrease in glomerular blood flow stimulates formation of Angiotensin II.
- (4) During summer when body loses lot of water by evaporation, the release of ADH is decreased.

180. Which one of the following statements are correct :-

- (1) The medullary zone of kidney is divided into a few conical masses called medullary pyramids projecting into the calyces
- (2) Inside the kidney the cortical region extends in between the medullary pyramids
- (3) Glomerulus alongwith Bowman's capsule is called the renal corpuscle
- (4) All the above



Space for rough work

Space for rough work

Space for rough work
